

From measurement to training: modelling learning with feedback for an EEG pattern-recognition certification test

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Abstract

The current CORTEX certification test is built on a signal-detection-theory (SDT) model in which each examinee is summarised by two parameters per task: a perceptual noise σ (sensitivity, $1/\sigma$ is the skill) and a decision criterion t (bias). Bayesian adaptive design (QUEST+-style) lets us *estimate* (σ, t) efficiently. In this note we summarise two internal design memos (`learning-theory1.md` and `learning-theory2.md`) that address the complementary problem: given $(\hat{\sigma}, \hat{t})$ for a learner, what is the optimal sequence of cases (with feedback) to actively *move* those parameters toward better values? The shift in viewpoint is that, in a training setting, σ and t are no longer stationary unknowns but *state variables*, and our job is to drive them. The design that emerges is a small, interpretable, posterior-driven policy that explicitly separates three training objectives, reuses the existing inference machinery, and exploits more structure than either the perceptual-learning or POMDP-teaching literatures use in isolation.

1 The conceptual shift: assessment vs. training

The existing CORTEX framework treats (σ, t) as fixed unknowns and selects each next case to maximise expected information gain about the joint posterior. That is provably efficient for *measurement*, but it is the wrong objective for *training*, where the learner's (σ, t) are precisely the things we want to change.

	Assessment	Training
(σ, t) are ...	stationary unknowns	state variables we want to move
Item-selection objective	expected information gain	expected improvement in future performance
Reward signal	posterior-variance reduction	$\Delta(- t)$, $\Delta(-\sigma)$, retention
Feedback role	none required	central; drives the dynamics

A measurement-optimal stimulus is typically placed *at* the criterion (uncertain by design); a training-optimal stimulus is typically placed at a different difficulty depending on which of three distinct learning sub-problems the learner currently has (Sec. 5).

2 Four (overlapping) literatures

Adaptive psychophysics and computerised adaptive testing. QUEST and QUEST+ (Watson & Pelli, 1983; Watson, 2017) and item-response-theory CAT both maximise information about a fixed ability parameter; neither models the parameter as changing. They are the right tools for our *measurement* sub-system and the wrong ones for a trainer.

Psychophysics of perceptual learning. Doshier and Lu’s Augmented Hebbian Reweighting Model and its successors give a mechanistic, trial-by-trial account of how repeated practice reduces σ (via channel reweighting, noise suppression, signal enhancement) and how feedback shifts t . The single most operationally useful empirical result, from Aberg & Herzog (2012) and earlier work by Herzog and colleagues, is that **block feedback primarily moves σ while trial-by-trial feedback primarily moves t** , and that biased trial-wise feedback can be used to drive t in a controllable direction.¹ The two parameters respond to different knobs and probably need different schedules.

The 85 % rule for optimal training difficulty. Wilson et al. (2019) derive that for a broad class of stochastic-gradient-descent binary classifiers — and argue this generalises to humans on perceptual binary-decision tasks — the *rate* of learning is maximised when training accuracy is held near 85% (error rate $\approx 15.87\%$). This is a formal derivation of Vygotsky’s zone of proximal development and Bjork’s “desirable difficulty,” and converges with established staircasing practice. Critically the 85% refers to *training* accuracy and *not* test accuracy — exactly the distinction this note is about.

Perceptual-learning systems in medical imaging. The closest existing analog to what we want to build is the Kellman group’s Perceptual-Adaptive Learning Module (PALM) family, scheduled by the ARTS algorithm (Adaptive Response-Time-based Sequencing, Mettler et al., 2011), which prioritises categories by per-category accuracy, response time as a fluency proxy, and trials-since-last-presentation. Kellman et al. (2024) is the first to integrate SDT into the pipeline: a running window d' serves as retirement criterion and beats accuracy-based retirement on both immediate and delayed post-tests. As far as we can tell this is the only existing perceptual-learning system that unifies measurement, sequencing, and mastery in a single framework, and we borrow liberally from its framing.

POMDP / Bayesian knowledge tracing. The decision-theoretic “right answer” to optimal teaching is Rafferty et al. (2016): cast teaching as planning in a partially observable Markov decision process whose hidden state is the learner’s knowledge state. This is the cleanest formal statement of our problem, but exact POMDP planning is heavy, and the standard formulations discretise mastery into binary states rather than maintaining a continuous posterior over (σ, t) . Spaced-repetition schedulers (SM-2, the Anki family) are simple heuristic approximations of this class and have strong empirical backing in medical education.

3 Why our problem has *more* structure than any of the above

The EEG-spike training problem is unusually well-conditioned because we already have:

1. a principled *generative model* of expert behaviour (a two-parameter probit with a lapse term);
2. a joint *posterior* over (σ, t) per examinee per task, maintained by the CORTEX engine;
3. stimuli that live on a known, signed *difficulty axis* s (the latent SDT signal), not in a discrete skill-tag space.

¹We do *not* propose to do this — see the caveats in Sec. 7.

This means learning can be expressed as *dynamics on the posterior-mean trajectory* (σ_k, t_k) rather than as discrete mastery transitions. Discrete-state BKT / POMDP work does not exploit this; the Kellman/ARTS priority score does not either. Our trainer can do better because the generative model is richer.

4 The POMDP formulation

The trainer is a partially observable Markov decision process (POMDP): we don't know the learner's true (σ, t) , we get only the binary response y_k each trial, and we have to choose the next stimulus s_k to maximise long-run improvement. This section writes that out carefully — first as a standard POMDP tuple, then giving explicit functional forms for the observation, transition, belief-update, reward, and policy components.

4.1 The POMDP tuple

A POMDP is a 7-tuple $(\mathcal{S}, \mathcal{A}, \mathcal{O}, T, Z, R, \gamma)$. For the trainer:

State $\mathcal{S}$.

$\theta_k = (\sigma_k, t_k, \phi_k)$. The perceptual noise $\sigma_k \in \mathbb{R}_{>0}$, the criterion $t_k \in \mathbb{R}$, and an optional vector ϕ_k holding nuisance state (lapse rate, sub-type-specific skill offsets, fatigue). *In testing θ is treated as stationary; in training it evolves under T .*

Action $\mathcal{A}$.

Choice of stimulus s_k for the next trial. In CORTEX the action space is discrete (pick one of the $\sim 10^4$ EEG segments in the bank, each indexed by its latent difficulty s on the signed SDT axis plus a categorical morphology / sub-type tag μ). For the math we treat the action as the pair $(s, \mu) \in \mathbb{R} \times \{1, \dots, M\}$.

Observation $\mathcal{O}$.

The learner's binary response $y_k \in \{0, 1\}$. Optionally an interaction trace (reaction time, number of answer changes, etc.) feeding ϕ_k ; we ignore that auxiliary channel below for clarity.

Transition kernel T .

$T(\theta_{k+1} \mid \theta_k, s_k, y_k, \text{fb}_k)$: the learning dynamics. Detailed in Secs. 4.3–4.4.

Observation kernel Z .

$Z(y_k \mid \theta_k, s_k)$: the SDT-with-lapse model. Detailed in Sec. 4.2.

Reward R .

$R(\theta_k, \theta_{k+1})$: improvement in bias, skill, and retention. Detailed in Sec. 4.6.

Discount γ .

$\gamma \in (0, 1]$. For a fixed-length training session of K trials, take $\gamma = 1$ and the planning horizon as the residual session length; for an open-ended trainer, $\gamma \lesssim 1$ to weight near-term improvement more.

Because θ_k is not directly observable, the trainer must act on its *belief state* b_k , the posterior over θ_k given all history through trial k . The POMDP reduces to an MDP on the infinite-dimensional belief simplex; in practice we represent b_k with the same SMC particle cloud the CORTEX engine already maintains.

4.2 Observation model: probit-lapse SDT

For stimulus s_k and learner state $\theta_k = (\sigma_k, t_k, \lambda_k)$ (we make the lapse rate λ explicit here),

$$Z(y_k = 1 \mid \theta_k, s_k) = (1 - 2\lambda_k) \Phi\left(\frac{s_k - t_k}{\sigma_k}\right) + \lambda_k, \quad (1)$$

identical to the CORTEX live-test engine. Φ is the standard normal CDF; the lapse mixture absorbs attentional / motor errors. This is the same model used during measurement — we are not re-deriving the observation channel, just naming it explicitly so the state dynamics below have something to interact with.

4.3 Transition kernel: criterion dynamics g_t

The criterion is updated each trial by a Rescorla–Wagner-style prediction-error rule. Let

$$\delta_k \equiv y_k - y_k^* \in \{-1, 0, +1\} \quad (2)$$

be the **outcome prediction error** on trial k , where y_k^* is the ground-truth label revealed by the feedback fb_k . Three cases:

- $\delta_k = +1$: false alarm. Learner said yes; correct answer was no. The criterion is too liberal — raise it.
- $\delta_k = -1$: miss. Learner said no; correct answer was yes. The criterion is too conservative — lower it.
- $\delta_k = 0$: correct. No update.

This gives

$$t_{k+1} = t_k + \alpha_t \delta_k + \xi_{t,k}, \quad \xi_{t,k} \sim \mathcal{N}(0, q_t^2) \quad (3)$$

i.e. Eq. (3) with the explicit form $g_t(s_k, y_k, \text{fb}_k) = \delta_k$.

Parameter glossary for Eq. 3.

$\alpha_t \geq 0$.

Criterion learning rate, in units of the perceptual axis (the same units as s and t). Estimated per-learner; empirical-Bayes prior from the pooled pilot cohort. Typical magnitudes from the SDT-criterion-learning literature (Stüttgen et al., 2011) place α_t between 0.05 and 0.2.

$\xi_{t,k}$.

Trial-level process noise with variance q_t^2 ; accounts for criterion fluctuation not captured by the deterministic R–W update (attention, mood, drift). Acts as the state-space-filtering noise that prevents the posterior from collapsing prematurely (see Sec. 4.5).

δ_k . As in Eq. 2: the only thing that drives t is being wrong, and the direction of the move is set by the direction of the error.

4.4 Transition kernel: skill dynamics g_σ

Skill (σ^{-1}) evolves much more slowly than criterion. The empirical learning curve in perceptual-learning experiments is well-approximated by an exponential approach to an asymptote — i.e. σ_k relaxes from σ_0 toward σ_∞ :

$$\sigma_k \approx \sigma_\infty + (\sigma_0 - \sigma_\infty) \exp(-k/\tau_\sigma). \quad (4)$$

The per-trial discretisation of this trajectory is a first-order relaxation:

$$\log \sigma_{k+1} - \log \sigma_k \approx -\frac{1}{\tau_\sigma} (\log \sigma_k - \log \sigma_\infty). \quad (5)$$

Two refinements turn this into a model we can drive with a policy.

(a) Difficulty-appropriateness weight. Not every trial contributes equally to skill. The 85 % rule (Wilson et al., 2019) says that trials at the *learner’s current* 85 % accuracy difficulty contribute most. Let

$$w(s_k, \theta_k) = \exp\left(-\frac{(|s_k - t_k|/\sigma_k - 1.04)^2}{2\rho^2}\right) \in (0, 1] \quad (6)$$

be a soft Gaussian indicator that peaks at $|s_k - t_k|/\sigma_k \approx 1.04$ (the 85 % point under unit-variance probit with lapse ≈ 0.025 , Eq. 15), with width ρ controlling how forgiving the criterion is. $w \approx 1$ for an ideally placed trial, $w \rightarrow 0$ for trials either far too easy (no challenge) or far too hard (no signal).

(b) Feedback-dependence. A trial without veridical feedback contributes negligibly to σ -learning (Aberg & Herzog, 2012). Let $f_k \in \{0, 1\}$ indicate whether immediate feedback was delivered on trial k .

Combining (a) and (b),

$$g_\sigma(s_k, \theta_k, \text{fb}_k) = -f_k \cdot w(s_k, \theta_k) \cdot (\log \sigma_k - \log \sigma_\infty), \quad (7)$$

yielding

$$\boxed{\log \sigma_{k+1} = \log \sigma_k + \alpha_\sigma g_\sigma(s_k, \theta_k, \text{fb}_k) + \xi_{\sigma,k}, \quad \xi_{\sigma,k} \sim \mathcal{N}(0, q_\sigma^2).} \quad (8)$$

Parameter glossary for Eq. 8.

$\alpha_\sigma \geq 0$.

Skill learning rate. With Eq. 7’s form, $\alpha_\sigma = 1/\tau_\sigma$ when $w = f = 1$. Per-learner; typical perceptual-learning τ_σ values in the literature are tens to a few hundred trials.

σ_∞ .

Asymptotic noise floor — the irreducible perceptual noise even after long training, set by the learner’s intrinsic sensory limits. Per-learner; weakly identified from a short training session, hence the empirical-Bayes prior.

ρ . Width of the 85 %-rule weight; how off-target a trial can be and still meaningfully contribute. Cohort-level nuisance.

$\xi_{\sigma,k}$.

Process noise on $\log \sigma$ with variance q_σ^2 . Same role as $\xi_{t,k}$ in Eq. 3.

What is *not* in g_σ . We omit (i) explicit dependence on the correctness $c_k = \mathbf{1}[y_k = y_k^*]$ — feedback delivers the correct answer either way, and the perceptual-learning literature suggests being wrong is not strictly necessary for σ -improvement — and (ii) sub-type-specific transfer terms, which can be folded in via ϕ_k once we have enough data to fit them. Both are natural extensions for Phase 3.

4.5 Belief state and the Bayesian filter

Let the belief at the start of trial k be $b_k(\theta) = p(\theta_k \mid \mathcal{H}_k)$, where $\mathcal{H}_k = \{s_j, y_j, \text{fb}_j\}_{j < k}$ is the history. After observing y_k and revealing fb_k , two operations update the belief:

Measurement step (Bayes on Z):

$$b_k^+(\theta) \propto Z(y_k \mid \theta, s_k) b_k(\theta). \quad (9)$$

Prediction step (push through T):

$$b_{k+1}(\theta') = \int T(\theta' \mid \theta, s_k, y_k, \text{fb}_k) b_k^+(\theta) d\theta. \quad (10)$$

Together, Eqs. 9–10 are a sequential Bayesian (state-space) filter — exactly the operation a Kalman filter performs in linear-Gaussian settings, generalised here to a nonlinear non-Gaussian state. The process noises $\xi_{t,k}, \xi_{\sigma,k}$ in Eqs. 3, 8 matter operationally: they prevent the posterior from concentrating prematurely, which would make the trainer unable to track real drift in (σ_k, t_k) .

In CORTEX the SMC particle representation makes this concrete: the existing N -particle cloud over (σ, t) is reweighted by Eq. 9, then each particle is propagated forward by sampling from T in Eq. 10; resampling whenever ESS drops below a threshold. The only engine change required is the addition of the propagation step — the reweighting step is already there.

4.6 Reward

A composite of the three things we actually care about:

$$R(\theta_k, \theta_{k+1}) = \beta_t(|t_k| - |t_{k+1}|) + \beta_\sigma(\sigma_k - \sigma_{k+1}) + \beta_r \cdot \text{ret}_k, \quad (11)$$

where ret_k is a retention term — for example, +1 for a successful spaced retrieval of a previously mastered sub-category, 0 otherwise. The weights $\beta_t, \beta_\sigma, \beta_r$ are chosen by the protocol designer based on what matters clinically (a small bias matters more than the marginal σ -reduction once σ is already low; spaced retrieval matters more once $|t|, \sigma$ are below clinical tolerance). Note all three terms are *differences* between consecutive states, so R is the state-improvement per trial.

4.7 Optimal policy: the Bellman equation and its approximations

The state-value function of belief state b under policy π is

$$V^\pi(b) = \mathbb{E}_\pi \left[\sum_{j=0}^{H-1} \gamma^j R_{k+j} \mid b_k = b \right], \quad (12)$$

where H is the remaining session horizon. The optimal value function V^* satisfies the Bellman equation

$$V^*(b) = \max_{s \in \mathcal{A}} \mathbb{E}_{y \sim Z(\cdot|b,s)} \left[\mathbb{E}_{\theta, \theta' \sim T,b} [R(\theta, \theta')] + \gamma V^*(b'(b, s, y)) \right], \quad (13)$$

where $b'(b, s, y)$ is the post-update belief produced by Eqs. 9–10. The optimal policy is $\pi^*(b) = \arg \max_s \{\dots\}$.

Exact solution of Eq. 13 is intractable on continuous (σ, t) . Three approximation tiers, in order of effort:

1. **One-step greedy (myopic) policy.** Drop the $\gamma V^*(b')$ term:

$$\pi_{\text{greedy}}(b) = \arg \max_s \mathbb{E}_{y, \theta, \theta'} [R(\theta, \theta')]. \quad (14)$$

Equivalent to a contextual bandit. Cheap; uses the existing posterior; no rollouts.

2. **Mode-conditional greedy** (what we propose in Sec. 6.2). Replace the single $\arg \max$ with three sub- $\arg$ -maxes, one per training objective, gated by a small set of cut-offs on $|\hat{t}_k|$ and $\hat{\sigma}_k$. Loses some expressivity vs. Eq. 14 but gains interpretability and avoids reward-weight sensitivity (Eq. 11’s β ’s).
3. **Short-horizon Monte Carlo rollout.** Approximate V^* by simulating L forward rollouts of length H under a base policy from the SMC posterior, averaging the discounted-return estimates. Closes the gap to a full POMDP solve when needed for Phase 3.

The concrete proposal in Sec. 6 is tier (2). Tier (1) is the natural ablation baseline; tier (3) is the upgrade path once we have a fitted dynamics model from the pilot data.

5 Three training sub-problems, three policies

The single most important conceptual point — one the existing literature has somewhat muddled — is that *what the trainer should do depends on what is wrong with the learner*. There are three distinct objectives, each with a different “optimal-stimulus” rule.

Bias correction (reduce $|t|$). *Fast, feedback-driven.* The Herzog/Aberg and recognition-memory literature (Han & Dobbins, 2009) shows that t is adjusted implicitly and rapidly in response to outcome feedback, even without the learner’s awareness. The right intervention is to train at stimuli near the learner’s current $\hat{t}_k$, with balanced sign, so that feedback is maximally informative about criterion placement. An unbalanced base rate would induce an opposite bias by accident.

Skill acquisition (reduce σ). *Slow, many trials, classically stimulus-specific.* This is where the 85% rule applies directly. The right intervention is to train at intermediate difficulty calibrated to the learner’s *current* psychometric function. For the unit-variance probit with lapse $\lambda \approx 0.025$ used in CORTEX, the $|s - \hat{t}_k|$ that produces $\approx 85\%$ correct is

$$|s - \hat{t}_k| \approx 1.04 \hat{\sigma}_k. \quad (15)$$

Diversity across morphologies matters here — Sec. 6.2 returns to this.

Retention and fluency. *Spaced, interleaved.* Spaced retrieval practice with feedback has strong support in medical education (Carpenter et al., 2022) and especially in pattern-recognition contexts (radiology and histopathology). The right intervention is interleaved return-visits to previously mastered sub-categories on a roughly exponential spacing schedule (SM-2 / Anki-family).

These three objectives demand different policies. A single-objective adaptive algorithm (QUEST+ for estimation, or a bare 85 %-rule staircase) will be wrong roughly two-thirds of the time.

6 A concrete proposal

The proposal is deliberately a small, interpretable policy that exploits the existing posterior rather than a full POMDP solve.

6.1 Learner-state model

Augment the existing (σ, t) generative model with the learning-dynamics layer of Eqs. (3)–(4). Estimate $\tau_\sigma, \sigma_\infty, \alpha_t$ per learner from their accumulated trial history; use empirical-Bayes priors derived from the pooled pilot cohort.

6.2 Item-selection policy

On trial k , with posterior means $\hat{\sigma}_k, \hat{t}_k$ in hand:

1. Choose the mode.

- *Bias-correction mode* if $|\hat{t}_k|$ is large relative to its posterior SD *and* above a clinical tolerance t^* .
- *Skill-building mode* otherwise, as long as $\hat{\sigma}_k$ is still decreasing (non-negligible learning-curve slope from Eq. 4) and above the target σ^* .
- *Retention mode* otherwise (skill plateaued, bias within tolerance).

2. Sample s_k accordingly.

- Bias mode: s_k near $\hat{t}_k$, with balanced sign.
- Skill mode: $|s_k - \hat{t}_k| \approx 1.04 \hat{\sigma}_k$ (Eq. 15); rotate through morphology sub-types for diversity.
- Retention mode: sample from previously seen difficulty bins on a spaced schedule.

3. Always give trial-wise veridical feedback.

The correct answer plus a brief feature-based rationale (annotations on the trace itself, in the style of the Kellman ECG PALM). The feature explanation turns the trial into a *learning-to-see* episode rather than a pure reinforcement episode.

This is not a full POMDP solve but it exploits all the structure the generative model provides — more than the standard adaptive-learning literature does.

6.3 Stopping / mastery criterion

Retire the learner from a task — or graduate them to maintenance-spacing — when, jointly,

$$\text{SD}(\sigma) \leq \delta_\sigma, \quad \hat{\sigma} \leq \sigma^*, \quad |\hat{t}| \leq t^*, \quad (16)$$

for pre-specified targets. This is a Bayesian upgrade of the Kellman/Massey “running- d' window” rule.

6.4 Suggested phased pipeline

1. **Phase 1 — Calibration.** 10–20 trials of the existing QUEST+ design to seed $(\hat{\sigma}_0, \hat{t}_0)$.
2. **Phase 2 — Adaptive training.** At each trial, update the posterior over (σ, t) , score candidate cases against Eq. 11, select per Sec. 6.2.
3. **Phase 3 — Learn the learning model.** From accumulated trial-by-trial data, fit a hierarchical model of how different cases change (σ, t) (identifying which cases reduce bias fastest, which improve discrimination, and any sub-type-specific learning effects). Use the fitted dynamics to upgrade the policy from heuristic to contextual-bandit, and eventually to a planned POMDP rollout.

7 Caveats

Stationarity assumption breaks during training. The existing CORTEX posterior-update implicitly assumes (σ, t) are fixed; during training they move. Three options: (a) reset the posterior per session and re-estimate from scratch; (b) explicitly model the dynamics and *filter* (sequential Bayesian state-space filtering) rather than just accumulate; (c) assume within-session stationarity and reset between sessions. Option (b) is the most principled.

Biased feedback is ethically off-limits. Herzog showed that biased trial-wise feedback shifts the perceived identity of a stimulus — explicitly *not* what we want in a clinical-skill trainer, where the learner must internalise the ground-truth label distribution. **Use veridical feedback only.** Influence t via the distribution of stimuli sampled and the presented base rate, never by lying about correctness.

Identifiability under joint motion. If the learner improves quickly, a drop in error rate could reflect either σ shrinking or t moving toward zero. The existing framework disentangles these via the stimulus-level response pattern, but it is worth verifying disentanglement still holds when both parameters move simultaneously. A small simulation study paralleling the existing measurement-side identifiability work would settle this.

Near-threshold is optimal for measurement, not learning. A trainer that simply re-uses the measurement-optimal stimulus placement will sit the learner at near-50 % performance — useful for estimating t , demoralising for learning. The 85 % rule and the Sec. 6.2 mode-switch exist precisely to avoid this trap.

8 Next steps

1. Extend the existing CORTEX simulation harness with the learning-dynamics model (Eqs. 3–4); verify joint identifiability of (σ_k, t_k) trajectories during simulated training.

2. Benchmark the three-mode policy against simpler baselines — method-of-constant-stimuli with feedback, a pure 85 %-rule staircase, a pure threshold-tracking adaptive scheme — on learning-curve metrics:
 - trials to reach target σ^* ,
 - trials to reach $|t| \leq t^*$,
 - post-test performance at held-out difficulty bins,
 - retention at $\Delta t = 7$ and 30 days.
3. Build a sibling app to CORTEX — same posterior, same engine — that runs the trainer policy and delivers feature-annotated feedback. Target form-factor: ~ 15 minutes per day, web or desktop bundle, same distribution story as the current pilot.
4. Validate against real expert data: do measured (σ, t) trajectories under the trainer actually follow the shapes Eqs. 3–4 predict? Re-fit and iterate.

Source documents

This note synthesises `learning-theory1.md` and `learning-theory2.md` from `GithubRepos/ideal-test/docs/`. Treat that pair as the primary source; this LaTeX version exists to be a single shareable design memo with consistent notation and mathematics.

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